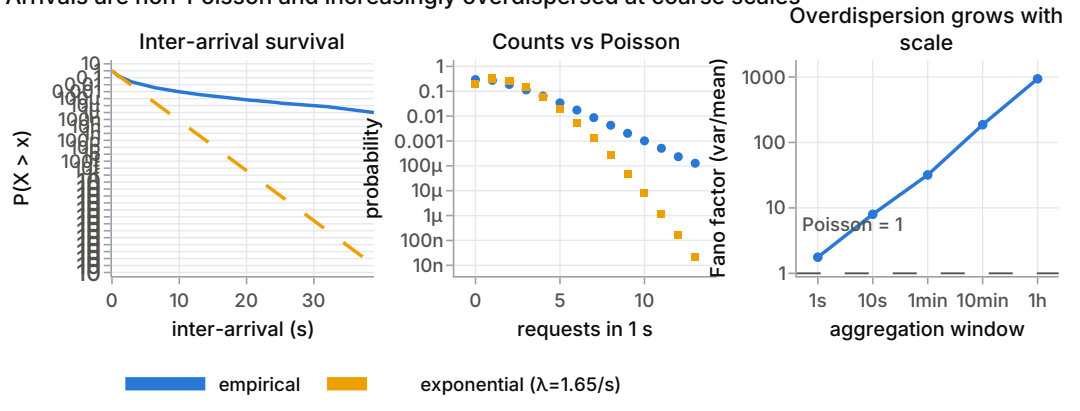Arrivals are non-Poisson and increasingly overdispersed at coarse scales